BTS-VIII(R)-04-23-1619 Reg. No. [ ][ ][ ][ ][ ][ ][ ][ ]

B

B.Tech. Degree VIII Semester Regular Examination April 2023

ME 19-205-0810 OPERATIONS RESEARCH
(2019 *Scheme*)

Time: 3 Hours Maximum Marks: 60

Course Outcomes
On successful completion of the course, the students will be able to:

CO1: Solve problems using optimization methods.
CO2: Develop mathematical skills to approach real life industrial problems.
CO3: Inculcate ideas to solve problems on Transportation and Queueing models.
CO4: Import and analyse case studies to solve future problems.
Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 –Analyze, L5 – Evaluate, L6 – Create
PO – Programme Outcome

PART A
(Answer *ALL* questions)

		$(8 \times 3 = 24)$	Marks	BL	CO	PO
I.	(a)	What do you mean by standard form of an LPP?	3	L2	1	1
	(b)	Write a note on assignment problem. Write the mathematical model of assignment problem.	3	L1	3	2
	(c)	What are the applications of LP in industry?	3	L3	1	2
	(d)	What are balanced and unbalanced transportation problem?	3	L1	3	1
	(e)	Differentiate basic, feasible, and optimum solutions in an LPP.	3	L3	2	2
	(f)	What is mean by unbounded solution?	3	L3	2	2
	(g)	Write a note on arrival pattern and service discipline in the case of queuing theory with the help of diagrams.	3	L1	4	3
	(h)	Discuss the feature of dynamic programming.	3	L2	4	2

PART B

$(4 \times 12 = 48)$

			Marks	BL	CO	PO
II.		A company produces chairs and tables. Wood and adhesive are required for the manufacturing of chairs and tables. 1 unit of wood and 3 units of adhesive are consumed in the manufacturing of each unit of the chair while it is 4 units and 1 unit respectively for the table. Suppose further that the company has 24 units of wood and 21 units of adhesives available per day, and that the net profits for selling units of the chair and table are given respectively as ₹2 and ₹5 per unit. How many chairs and tables they should produce to maximize the profit? Formulate as an LPP and solve using graphical method.	12	L3	1	2

OR

			Marks	BL	CO	PO
III.	(a)	Discuss the significance and scope of OR in modern management.	3	L2	1	1
	(b)	Solve using Graphical method.	9	L5	1	3

Maximize: $Z = 20x + 10y$

Subject to: $x + 2y \leq 28$

$3x + y \leq 24$

$x \geq 2$

$y \geq 0$

(P.T.O.)

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			Marks	BL	CO	PO
IV.	(a)	State any two methods of solving LPP.	2	L1	2	2
	(b)	Solve using simplex method Max $Z = 6x + 3y$ Subject to: $2x + 5y \leq 120$ $2x + y \leq 40$ $x, y \geq 0$	10	L5	2	2

OR

			Marks	BL	CO	PO
V.		Solve using Big M method Maximize: $Z = x + 2y$ Subject to the constraints: $x - y \geq 3$ $2x + y \leq 10$ $x \geq 0, y \geq 0$	12	L5	2	2
VI.		Solve the following transportation problem and check for optimality using MODI or stepping stone method.	12	L4	3	2

Orgin/Dest	D1	D2	D3	ai
O1	2	7	4	5
O2	3	3	1	8
O3	5	4	7	7
O4	1	6	2	14
bj	7	9	18	34

OR

			Marks	BL	CO	PO
VII.	(a)	Explain degeneracy in a transportation problem.	3	L3	3	1
	(b)	Solve the following assignment problem for assigning sub-assemblies to contractors to minimize the cost.	9	L5	3	2

Contractor / Sub Assembly	C1	C2	C3	C4
S1	15	13	14	17
S2	11	12	15	13
S3	13	12	10	11
S4	15	17	14	16

			Marks	BL	CO	PO
VIII.	(a)	Define Bellman's principle of optimality.	2	L2	4	2
	(b)	A salesman wants to go from city 1 to city 10. The various possibilities and distances(in km) are given in the following table. Use dynamic programming for solving the problem.	10	L5	4	2

Arc	Distance	Arc	Distance	Arc	Distance	Arc	Distance
1-2	5	2-7	8	4-6	5	6-9	7
1-3	5	3-5	8	4-7	7	7-8	5
1-4	6	3-6	10	5-8	6	7-9	7
2-5	4	3-7	5	5-9	8	8-10	8
2-6	7	4-5	4	6-8	9	9-10	9

OR

			Marks	BL	CO	PO
IX.	(a)	Write down the cost structure of a queuing system.	4	L2	4	2
	(b)	A self-service store employs one cashier at its counter. 8 customers arrive on an average every 5 minutes while the cashier can serve 10 customers in the same time. Assuming poisson arrival and exponential distribution for service Find: (i) Average number of customers in the system. (ii) Average time a customer waits before being served. (iii) Probability that there is no customer in the system. (iv) Probability that there are more than two customers in the system. (v) Probability that the server is idle.	8	L5	4	2

Bloom's Taxonomy Levels
L1 - 19%, L2 - 24%, L3 – 24%, L4 – 4%, L5 – 29%.

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